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Safety and Efficacy of Ianalumab in Patients With Sjögren's Disease: 52-Week Results From a Randomized, Placebo-Controlled, Phase 2b Dose-Ranging Study

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Objective. The objective of this study was to report 52-week safety and efficacy of ianalumab from phase 2b dose-finding study in patients with Sjögren's disease (SjD).

Methods. Patients randomly received (1:1:1:1) ianalumab (5, 50, or 300 mg) or placebo subcutaneously every 4 weeks until week 24 (treatment period [TP]1). At week 24, patients on 300 mg were rerandomized to continue 300 mg or receive placebo until week 52 (TP2), patients on placebo were switched to ianalumab 150 mg, and patients on 5 and 50 mg directly entered posttreatment safety follow-up. Patients who discontinued treatment early or completed treatment entered safety follow-up (≥20 weeks).

Results. During TP1, 190 patients were randomized (placebo = 49, 5 mg = 47, 50 mg = 47, 300 mg = 47). Of these 190 patients, 90 (47.4 %; 43 continued 300 mg and 47 received placebo) entered TP2, and 81 of 90 (90.0%) completed the study treatment. By week 52, efficacy was sustained in patients who continued 300 mg in TP2 (EULAR Sjögren's Syndrome Disease Activity Index, EULAR Sjögren's Syndrome Patient Reported Index, patient global assessment, and physician global assessment change from week 24: −1.45, −0.46, −4.69, and −6.86, respectively). Stimulated salivary flow rates and autoantibody levels numerically improved in the 300 mg group. Treatment-emergent adverse events were not dose-dependent, except for injection-site reactions. Cases of decreased neutrophil counts (Common Terminology Criteria for Adverse Events v4.03 grade 3 according to laboratory listings) were observed in three patients during the posttreatment follow-up, occurring at 3.5, 5.5, and 3 months, after the last ianalumab administration. None were associated with infection except one incidental finding of asymptomatic cytomegalovirus infection (IgM-positive).

Conclusion. In patients with SjD, ianalumab 300 mg demonstrated sustained efficacy through week 52 and a favorable safety profile up to two years of follow-up.

INTRODUCTION

Sjögren's disease (SjD) is an autoimmune disease characterized by lymphocytic infiltration, malfunction, and progressive destruction of exocrine glands. The clinical features range from mucosal dryness, fatigue, and pain, affecting nearly all patients, to severe extraglandular disease manifestations with an increased risk for lymphoma development in a subset of patients.[1] The

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treatment goals in patients with SjD are alleviation of symptoms of dry eye and mouth, control of systemic manifestations, and improvement in quality of life.[2] Currently available treatments for SjD are limited to symptomatic care for mucosal signs and symptoms.[3,4] Although conventional disease-modifying antirheumatic drugs and glucocorticoids have been shown to reduce severe inflammation and delay the progression of systemic disease, they are often prescribed for off-label use and reported to be ineffective in improving the symptoms of dryness and fatigue.[5] Over the past two decades, a number of disease-modifying treatments targeting specific elements of the immune system have been evaluated in SjD.[2]

Recent advances in the understanding of the pathogenesis of SjD have shown that B cells play a key role in the disease development and progression.[6,7] Biologic therapies directed against cellular B cell targets, including rituximab and epratuzumab, or against the soluble cytokine B cell activating factor (BAFF) (belimumab) have not shown potential benefits in treating patients with SjD.[8–13] A meta-analysis of rituximab trials showed weak effects on salivary and tear flow but no improvement in fatigue or other global measures of well-being.[8–10] Treatment with epratuzumab improved symptoms only in some patients in an open-label phase 1/2 study[11] and in a subgroup of patients with systemic lupus erythematosus (SLE) with associated SjD.[14] Belimumab improved fatigue in open-label phase 2 studies but was not effective in improving symptoms of dryness.[12,13] The combination of rituximab and belimumab has been shown to enhance the depletion of salivary gland B cells and improve clinical outcomes.[15] However, available data are limited to a single randomized controlled SjD trial, and further studies are needed to delineate the benefit and risk of this approach.

Ianalumab (VAY736) is a novel B cell–targeting human monoclonal antibody that combines profound B cell depletion via antibody-dependent cellular cytotoxicity with blockade of the BAFF receptor (BAFF-R), interrupting BAFF-mediated signaling for B cell maturation, proliferation, and survival, and is hypothesized to prevent reactivation of residual pathogenic B cell clones in the target tissue. In this paradigm, suppression of these B cell clones despite continued BAFF upregulation may play a role in the long-term maintenance of the treatment response to ianalumab after initial cell depletion.

A phase 2b dose-finding study evaluated three dose levels of ianalumab versus placebo in patients with moderate to severe SjD.[16] The study met its primary end point at week 24; ianalumab treatment resulted in improvements in the EULAR Sjögren's Syndrome Disease Activity Index (ESSDAI), demonstrating a statistically significant dose-response relationship. Ianalumab was well tolerated with no dose-dependent increase in infections until week 24. Trends in other outcome measures and significant improvements in stimulated salivary flow further support the findings of a positive risk-benefit of ianalumab in this trial.[16] Herein, we report the 52-week safety and additional exploratory efficacy results of this study.

PATIENTS AND METHODS

Study design and patients. This study was a randomized, parallel, double-blind, placebo-controlled, phase 2b multicenter (65 sites in 19 countries) dose-finding trial of ianalumab in patients with SjD. The detailed study design and patient inclusion and exclusion criteria have been previously reported.[16] Briefly, patients aged 18 to 75 years who fulfilled the American European Consensus Group classification criteria[17] were included. In addition, eligible patients were positive for anti-Ro/SSA antibody, had an ESSDAI score of 6 or more at screening (eligibility score based on weighted scores of the biologic, hematologic, articular, cutaneous, glandular, lymphadenopathy, and constitutional domains), and had a EULAR Sjögren's Syndrome Patient Reported Index (ESSPRI) score of 5 or more at baseline and stimulated whole salivary flow rate of more than 0.1 mL/min (minimal level of saliva production) at screening.

Key exclusion criteria were evidence of SjD associated with other autoimmune diseases, serious diseases or uncontrolled conditions, active infection, malignancy within the past five years (other than localized basal cell carcinoma of the skin, in situ cervical cancer or SjD-related lymphoma), pregnancy, recent change in allowed background therapy, and prior exposure to biologics within 180 days before randomization. After a screening period of four weeks, eligible patients were randomized equally to receive subcutaneous (SC) doses of ianalumab (5, 50, or 300 mg) or placebo every four weeks up to week 24. Per the study design, at week 24, after completion of the first blinded treatment period (TP)1, patients on ianalumab 300 mg were rerandomized (1:1) to continue 300 mg or receive placebo every 4 weeks up to week 52 in TP2, and patients on placebo were switched to ianalumab 150 mg every 4 weeks up to week 52. Patients in the ianalumab 5 and 50 mg groups directly entered safety follow-up. Patients who prematurely discontinued the study treatment at any time point or completed the treatment as planned entered the safety follow-up period for ≥20 weeks posttreatment (Supplementary Figure 1). The duration of safety follow-up was up to two years from the last dose of the study treatment, and it was defined for each patient based on the level of recovery of CD19–positive (CD19+) B cells.

All patients provided written informed consent before inclusion in the study. The trial was conducted according to Good Clinical Practice and the Declaration of Helsinki. The review board at each site approved the protocol. Data obtained at each site were monitored and analyzed by Novartis personnel.

Outcomes and assessments. As previously reported,[16] the primary outcome was the change in the ESSDAI score from baseline at week 24. The exploratory outcomes up to week 52 included changes from baseline in ESSDAI, ESSPRI, physician global assessment (PhGA), patient global assessment (PaGA), Functional Assessment of Chronic Illness Therapy Fatigue

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(FACIT-F), 36-Item Short Form Health Survey (SF-36) scores, and change from baseline in the levels of soluble BAFF, IgG, rheumatoid factor (RF), and autoantibody (anti-SSA/Ro60 and 52, anti-SSB/La).

The secondary outcomes related to objective measures included (1) change from baseline in stimulated salivary flow at week 24 and (2) change from baseline in whole-blood $CD19^+$ B cell counts and time to recovery defined as ≥80% of baseline or ≥50 cells/μL.

Regular safety monitoring was performed: patient on-site visits occurred every four weeks and included a physical examination, routine laboratory tests, and evaluation and collection of adverse event (AE) details. Serious AEs (SAEs) were reported upon occurrence. The AE severity was graded as low, moderate, or severe. Clinical efficacy measurements, including ESSDAI and PhGA, and patient-reported outcome (PRO) measures, ESSPRI, SF-36, FACIT-F, and PaGA were collected at baseline and every four weeks during TP1 and TP2. Stimulated salivary flow was assessed at baseline and at weeks 12, 24, 28, 36, and 48.

Statistical analysis. The full analysis set (FAS) comprised all randomized patients to whom the study treatment was assigned. FAS was used for all efficacy variables unless otherwise stated. The safety set included all patients who received at least one dose of the study medication. The safety set was used in the analysis of all the safety variables. Demographics and baseline characteristics were descriptively summarized for the treatment groups using frequencies, percentages, or means (SD), as appropriate. For the secondary variables, ESSPRI and FACIT-F score dose-response analysis using Multiple Comparison Procedure–Modeling was performed, and all other secondary end points were analyzed using a mixed-effect model for repeated measures. Descriptive statistics were provided for ESSDAI, ESSPRI, FACIT-F, PhGA, PaGA, and SF-36. Continuous efficacy data were analyzed as the change from baseline with a linear mixed-effect model for repeated measures. The percentage change from baseline in B cell counts before, during, and after treatment with ianalumab, as well as the time to recover to baseline-like values (defined as at least 80% of baseline counts or ≥50 cells/μL), were descriptively summarized. Post hoc analysis of the relationship between baseline BAFF concentrations and ESSDAI response at week 24 was performed using the one-sided *t*-test.

RESULTS

Patient disposition and characteristics. Of the 190 patients randomized in TP1 (placebo, n = 49; ianalumab 5 mg, n = 47; ianalumab 50 mg, n = 47; ianalumab 300 mg, n = 47), 178 (93.7%) completed TP1 as planned. As per protocol, 47 patients each in the ianalumab 5 and 50 mg groups entered the safety follow-up period directly from TP1.

A total of 90 of 190 (47.4%; ianalumab 300 mg, n = 43; placebo, n = 47) patients entered TP2.[16] At week 24, patients who received ianalumab 300 mg were rerandomized to the placebo (n = 22) or ianalumab 300 mg (n = 21) group, and patients in the placebo group were switched to ianalumab 150 mg (n = 47). A total of 81 patients (90.0%) completed the planned study treatment in TP2, and 9 (10%) discontinued prematurely. The most common reason for discontinuation was AEs (n = 7; 7.8%). A total of 88 patients (placebo to ianalumab 150 mg, n = 46; ianalumab 300 mg to placebo, n = 22; ianalumab 300 mg to ianalumab 300 mg, n = 20) from TP2 continued to the safety follow-up period.

Demographics and baseline clinical characteristics were comparable between groups in TP2 (Table 1). Overall, the mean ESSDAI score was ≥10 for more than 70% of patients in each treatment group, with a mean ESSDAI score of 13.5 in the ianalumab 300 mg to ianalumab 300 mg group.

Efficacy. *Primary outcome.* As reported previously, the primary end point was met[16]; statistically significant dose-related decrease in disease activity, measured by ESSDAI, was observed at week 24. Briefly, the largest ESSDAI improvement was 1.92 points more reduction with ianalumab 300 mg versus placebo at week 24.

Exploratory and secondary outcomes. For TP2, one of the main objectives was to descriptively evaluate differences within the patient group on 300 mg who either discontinued or continued ianalumab 300 mg by week 24. At week 52, efficacy was sustained for patients who continued ianalumab 300 mg (n = 21) in TP2 (ESSDAI, ESSPRI, PaGA, and PhGA change from week 24: −1.45, −0.46, −4.69, −6.86, respectively; Supplementary Table 1). By week 52, improvement was also noted for patients in the placebo group who switched to ianalumab 150 mg. At the same time point, some loss of week 24 improvements in PROs (FACIT-F, PaGA, and SF-36) was observed in patients who switched to placebo at week 24. The changes between weeks 24 and 52 were numerically smallest for ESSPRI. Figure 1 shows a summary of the key exploratory efficacy parameters at week 52 compared to week 24.

Biomarker analysis. A greater maximum reduction in $CD19^+$ B cells and IgG levels was observed at week 24 for the 50 mg and 300 mg groups compared to the 5 mg group (Figure 2 A–C). Across all treatment groups, BAFF sharply increased from baseline and promptly decreased after treatment discontinuation (Figure 2A). In the placebo to ianalumab 150 mg group, BAFF remained at baseline levels until week 24 and showed a respective increase upon the start of ianalumab after week 24, which was sustained until week 48 (last testing time point).

Per protocol, in TP1, the number of $CD19^+$ cells was measured before the first dose of the study treatment (baseline) and at the week-24 visit. At week 24, the mean B cell counts remained at the baseline level in the placebo-treated group, whereas all

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Table 1. Baseline demographic and clinical characteristics of patient subgroups treated in treatment period 2*

Variables	Placebo to ianalumab 150 mg (n = 47)	Ianalumab 300 mg to placebo (n = 22)	Ianalumab 300 mg to ianalumab 300 mg (n = 21)
Age, y	47.6 (12.6)	48.7(16.8)	48.7 (14.7)
Female, n (%)	45 (95.7)	21 (95.5)	21 (100)
ESSDAI	12.9 (6.7)	13.7 (7.1)	13.5 (6.7)
ESSPRI	7.3 (1.1)	6.7 (1.9)	6.9 (1.5)
FACIT-F score	23.2 (9.1)	28.3 (11.5)	26.9 (10.5)
PhGA	51.8 (16.9)	53.6 (14.3)	51.4 (15.8)
PaGA	61.6 (17.4)	68 (24.0)	56.5 (18.8)
Stimulated salivary flow rate (mL/min), [min–max]	0.54 (0.74) [0.0–3.7]	0.92 (1.13) [0.0–5.1]	0.70 (0.58) [0.0–1.9]
Schirmer's test: left eye, mm	7.3 (8.8)	10.4 (11.7)	7.6 (8.1)
Schirmer's test: right eye, mm	6.6 (6.8)	8.6 (9.9)	5.7 (7.6)
Positive anti-Ro/SSA status, n (%)	47 (100)	22 (100)	21 (100)
IgG, g/L	17.4 (7.1)	18.0 (6.7)	17.7 (8.6)
IgM, g/L	1.7 (1.5)	1.3 (0.8)	1.3 (1.2)
B cell count, cells/μL	199.6 (146.8)	265.2 (267.1)	278.5 (156.2)
Use of DMARDs, n (%)	27 (57.4)	12 (54.5)	10 (47.6)
Hydroxychloroquine, n (%)	25 (53.2)	11 (50.0)	8 (38.1)
Methotrexate, n (%)	4 (8.5)	5 (22.7)	2 (9.5)
Steroid therapy, n (%)	14 (29.8)	5 (22.7)	7 (33.3)

* Values are in mean (SD), unless specified otherwise. Baseline medications were taken before the first dose and continued through the treatment period with a missing stop date or with a stop date on or after the first dose. DMARD, disease-modifying antirheumatic drug; ESSDAI, EULAR Sjögren's Syndrome Disease Activity Index; ESSPRI, EULAR Sjögren's Syndrome Patient Reported Index; FACIT-F, Functional Assessment of Chronic Illness Therapy Fatigue; PaGA, patient global assessment; PhGA, physician global assessment.

ianalumab-treated patients showed a sharp decline. However, there was a greater maximum reduction in mean CD19 B cell counts observed for 50 and 300 mg than for 5 mg. Following a treatment switch from placebo to ianalumab 150 mg, a sharp drop in B cell counts was observed at week 28. The B cell counts remained suppressed in all treatment groups up to week 48; depletion was maintained in patients who continued to receive ianalumab 300 mg up to week 48, and a marked return of B cell counts was observed after week 36 in patients who switched from ianalumab 300 mg to placebo at week 24. The time to $CD19^+$ B cell recovery after stopping the study treatment was dose-proportional; the median time to B cell recovery was 3.8 months for ianalumab 5 mg, 4.8 months for ianalumab 50 mg, 8.4 months for ianalumab 300 mg (24 weeks), 6.8 months for placebo to ianalumab 150 mg (28 weeks), and 6.5 months for ianalumab 300 mg (52 weeks) groups (Supplementary Figure 2).

A reduction in IgM RF and serum IgG levels was observed across all treatment groups. The anti-Ro60/52 and anti-La autoantibody levels demonstrated a reduction of 15% to 20% from baseline at week 24 in the 50 and 300 mg groups. Notably, the decrease in anti-Ro60/52 levels was less prominent in the 5 mg group. Anti-Ro60/52 and anti-La autoantibody levels returned to baseline 6 months after discontinuation of ianalumab 300 mg. In contrast, continued treatment with ianalumab 300 mg showed stable or further decline in autoantibody levels (Figure 2D–G).

Other secondary outcomes. The stimulated whole salivary flow rate improved from baseline over time. The highest increase was observed with ianalumab 300 mg at week 24,[16] rising by 0.20 mL/min above placebo (95% confidence interval [CI]: 0.01–0.38; $P = 0.037$). At the last available assessment time point of flow rates (week 48), further improvement was observed in the ianalumab 300 mg group; stimulated salivary flow rate improvements remained numerically higher in the ianalumab 300 mg continued treatment group compared to the group that switched to placebo at week 24 (least squares mean change from baseline: 0.45 mL/min versus 0.22 mL/min; Figure 3). Supplementary Table 2 shows the proportion of patients with stimulated salivary flow response rate of at least 0.25 mL/min, and 0.50 mL/min over baseline. Figure 4 shows the potential associations between baseline BAFF levels with week 24 ESSDAI response in patients receiving an efficacious dose (50 and 300 mg pooled; analysis done for TP1 only). In this post hoc analysis, no statistical difference was observed between ESSDAI responders and nonresponders at the group level.

Safety. Safety data are presented for TP1, TP2, and post-treatment follow-ups. The 150 mg group includes TP1 placebo-treated patients who switched to ianalumab 150 mg at week 24 and "any 300 mg" group includes patients treated with ianalumab 300 mg up to week 24 plus those who continued 300 mg to week 52. The "any ianalumab"–treated group includes patients who received ianalumab treatment in TP1 or TP2.

Overall, a total of 176 of 188 patients (93.6%) who received ianalumab had at least one AE. The most frequent treatment-emergent AE (TEAE) per system organ class was "infections and infestations" (73.4%) followed by "general disorders and

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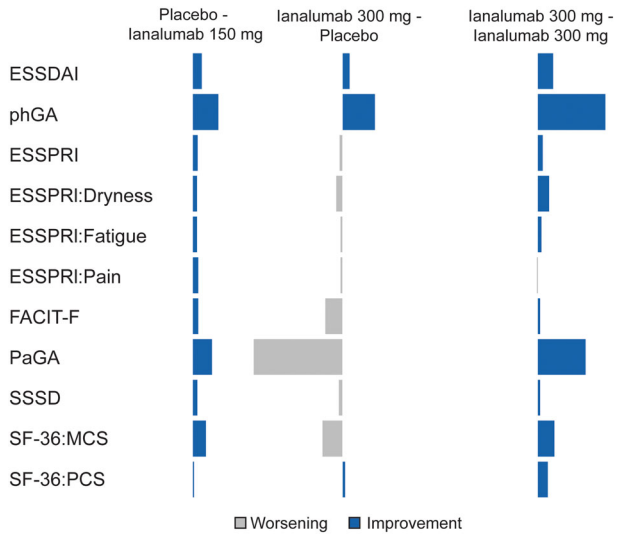

Figure 1. Summary of efficacy data at week 52 compared to week 24. Only patients with week 24 measurements and at least one measurement after week 24 were included. Least square means were derived from a mixed model for repeated measures that included treatment, visit, treatment by visit interaction, stratification factor, baseline ESSDAI score, and region as factors, and week 24 measurement as a continuous covariate. ESSDAI, EULAR Sjögren's Syndrome Disease Activity Index; ESSPRI, EULAR Sjögren's Syndrome Patient Reported Index; FACIT-F, Functional Assessment of Chronic Illness Therapy Fatigue; MCS, mental component summary; PCS, physical component summary; PaGA, patient global assessment; PhGA, physician global assessment; SSSD, Sjögren's syndrome symptom diary; SF-36, 36-Item Short Form Health Survey.

administration site conditions" (43.1%). Overall, a higher proportion of TEAEs was observed in the groups treated with ianalumab 300 mg, and these results were driven by mild to moderate injection-site reactions. One patient in the ianalumab 150 mg group reported a severe injection-site reaction and discontinued the study treatment. Overall, infections were mostly mild or moderate. The most commonly reported infections were nasopharyngitis (16.5%), upper respiratory tract infections (17.6%), urinary tract infections (13.8%), sinusitis (9.6%), bronchitis (8.0%), oral herpes (6.4%), and conjunctivitis (6.4%). A summary of AEs by treatment groups is provided in Table 2.

Serious infections were reported in eight patients treated with any ianalumab dose. The five SAEs of infections (Supplementary Table 3) were considered by the investigators to be treatment-related: candida infection (n = 1 in 5 mg group), tubo-ovarian abscess plus appendicitis (n = 1 in 50 mg group), and bronchitis and respiratory syncytial virus bronchiolitis (n = 1 in 150 mg group, in a patient with interstitial lung disease). The other three SAEs were not considered to be treatment-related: pneumonia (1 community-acquired, associated with chronic obstructive pulmonary disease in 150 mg group and 1 associated with newly diagnosed sarcoidosis in 300 mg group), wound infection (n = 1 in 50 mg group), sinusitis (n = 1 in 150 mg group), and COVID-19 and COVID-19 pneumonia (n = 1 in 300 mg group; both occurred 740 days after the last dose of ianalumab).

Malignancies were reported in two patients. A patient with previously treated human epidermal growth factor receptor–positive breast cancer was considered cured approximately 10 years previously and was randomized to ianalumab 5 mg. Nine months later, the patient had chest pain, and recurrence of breast cancer was suspected (treatment-emergent SAE). Positron emission tomography scan showed a rib fracture that was considered traumatic in nature. A course of trastuzumab treatment was initiated. No biopsy was performed; therefore, no tissue confirmation of the suspicion of recurrence of metastatic breast cancer was available. Based on the information obtained in April 2021, no further details related to the treatment and outcome of this case was available. The second malignancy case was deterioration of the concurrent extranodal marginal zone B cell lymphoma (mucosa-associated lymphoid tissue [MALT] type, SjD associated). One month before the start of study treatment with placebo, the patient experienced recurrence of the MALT lymphoma. Although further deterioration of lymphadenopathy was reported in TP2 (after the patient was rerandomized to ianalumab 150 mg), it was deemed unrelated to the study treatment by the investigator, and the treatment was continued. Some improvement of the lymphoma condition was noted three months later.

A total of 11 of 188 patients (5.9%) discontinued the study treatment due to AEs: 2 of 47 patients in the 5 mg group for bronchitis and decreased lymphocyte count; 2 of 47 patients in the 50 mg group for wound infection and injection-site reaction; 5 of 47 patients in the 150 mg group for herpes zoster, esophagitis, rash and vasculitis (occurred in the same patient), injection-site reaction, and systemic injection–related reaction; 2 of 21 patients in the 300-mg group for (1) pulmonary sarcoidosis and (2) arthralgia and dermatosis.

Lymphopenia and neutropenia were mostly grade 1 and grade 2. Cases of decreased neutrophil counts (Common Terminology Criteria for Adverse events [CTCAE] v4.03 grade 3 according to laboratory listings) were observed in three patients during the posttreatment follow-up, occurring at 3.5 months, 5.5 months, and 3 months after the last administration of ianalumab. All three patients had ongoing history of neutropenia CTCAE grade 1 (2 patients) or CTCAE grade 2 (1 patient) documented at screening or baseline visit. None were associated with infection except for a single incidental finding of an asymptomatic viral infection (cytomegalovirus [CMV] IgM-positive). Two patients had pre-existing grade 1 neutropenia, whereas one patient had pre-existing grade 2 neutropenia. No grade 4 neutropenia was reported. Two patients (n = 1 each in the 5 mg and 50 mg groups) entered the study (screening/baseline) with IgG levels below lower limit of normal (LLN), and their IgG levels remained below LLN until end of the study.

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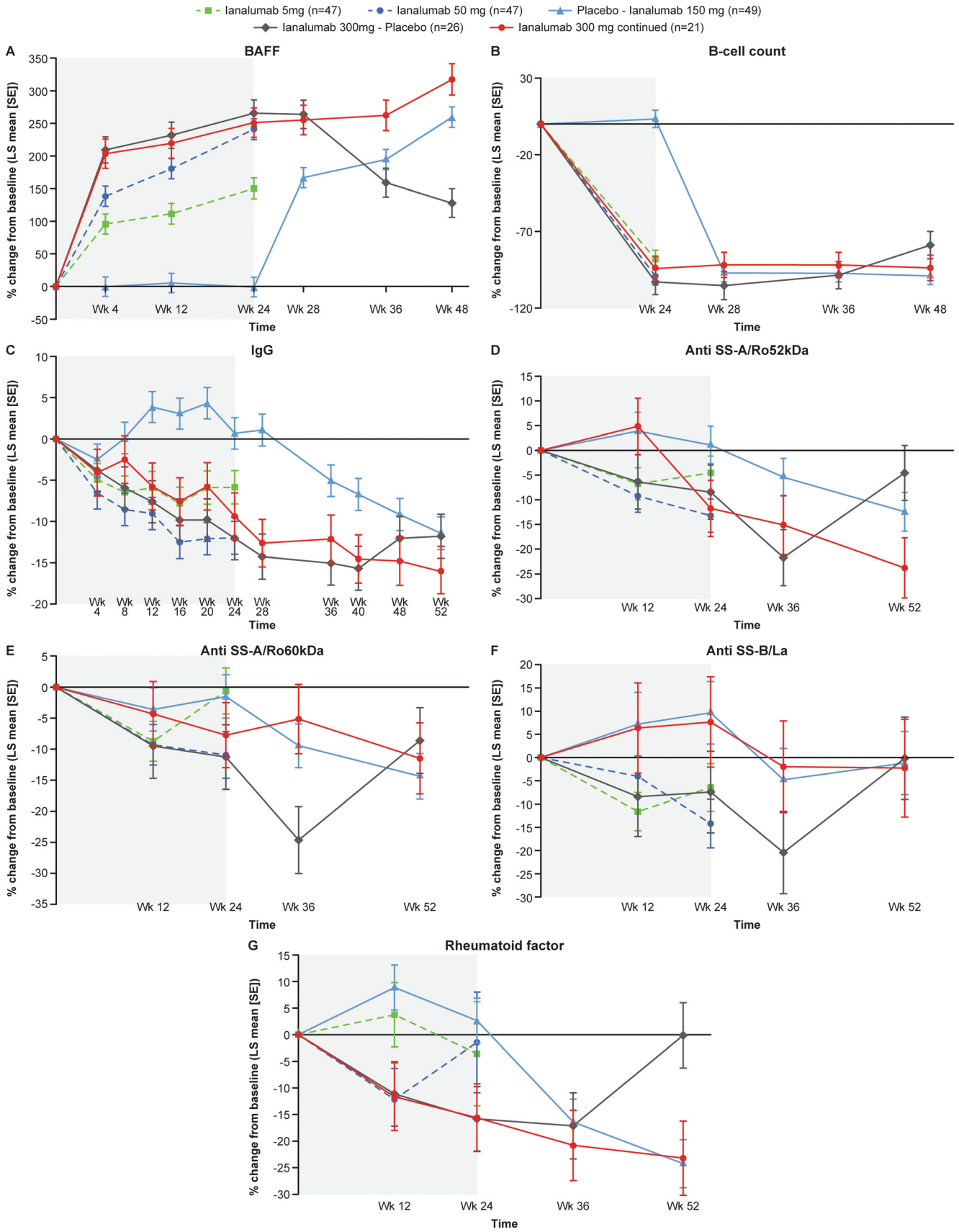

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Figure 2. Serum BAFF, B cells, IgG, and autoantibody concentrations by treatment group. The gray shaded area shows the placebo-controlled period. Patients who received 5 and 50 mg of ianalumab proceeded directly to the follow-up period. Therefore, no measurements were available for these cohorts after week 24. For (D)–(G), n = 47 for the placebo to ianalumab 150 mg and n = 22 for the ianalumab 300 mg to placebo groups. (A) LS mean change in BAFF concentration from baseline (pg/mL); (B) LS mean change in B cell count from baseline (cells/μL); (C) LS mean change in IgG concentration from baseline (g/L); (D–G) LS mean ± SE of change from baseline of biomarkers presented up to week 52. BAFF, B cell activating factor; LS, least squares.

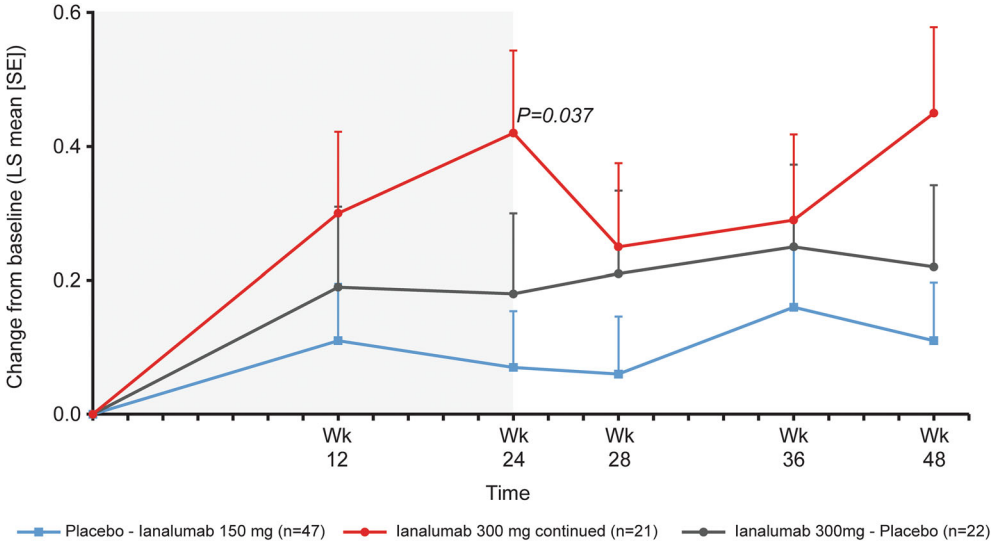

Figure 3. Stimulated salivary flow rate over time up to week 48, the last testing time point (primary end point at week 24). The gray shaded area shows the placebo-controlled period. Stimulated whole salivary flow significantly increased with 300 mg at week 24, rising by 0.20 mL/min above placebo (95% CI: 0.01–0.38; $P = 0.037$). Per protocol, no formal significance testing was performed for the exploratory time points after week 24. The figure presents data as the change from baseline for dose arms studied in TP2 following the new treatment assignment at week 24. Baseline is defined as the last assessment performed on or before the date of administration of the first dose of the study treatment. Only patients with baseline measurements and at least one measurement postbaseline were included. LS means were derived from a mixed model for repeated measures that included treatment, visit, treatment by visit interaction, stratification factor, baseline ESSDAI score, and region as factors and baseline value as a continuous covariate. CI, confidence interval; ESSDAI, EULAR Sjögren's Syndrome Disease Activity Index; LS, least squares; TP, treatment period.

DISCUSSION

As previously reported,[16] the phase 2b study of ianalumab in moderate to severe active SjD achieved its primary objective, showing a dose-dependent reduction in disease activity at week 24. A dose of 300 mg administered once a month showed the greatest improvement in several outcome measures.[16] The strength of the study design was the blinded prospective evaluation of efficacy with or without continuation of the investigational drug after the first blinded treatment period, between weeks 24 and 52. These exploratory efficacy results are the focus of the present paper, complemented by a detailed report of the final safety profile of ianalumab in SjD, including up to two years of drug-free follow-up.

After six months of treatment with the highest dose of ianalumab 300 mg every four weeks, patients were rerandomized to either discontinue or continue treatment with the same dose up to the end of TP2 at week 52. Although the results from physician-reported outcome instruments mostly remained stable, results from PRO instruments suggested an increase of disease activity approximately six months after discontinuation of ianalumab 300 mg. At the molecular level, IgM RFs, anti-SSA/Ro52 and 60, and anti-SSB/La autoantibody levels returned to baseline concentrations in patients who discontinued ianalumab, whereas continued ianalumab treatment showed sustained autoantibody responses or further declines. The differences in anti-La reductions between the two patient groups, both receiving 300 mg, from week 24 to 52 was unexpected and could be due to random variation and the small sample size. In summary, these insights support the notion that molecular changes related to B cell depletion or the BAFF/BAFF-R axis and reductions in autoantibody concentrations are related to the clinical benefit experienced by patients with SjD.

In line with the cellular and molecular mechanisms of action of ianalumab, a dose-response relationship for the pharmacodynamic markers serum BAFF and $CD19^+$ B cell count was

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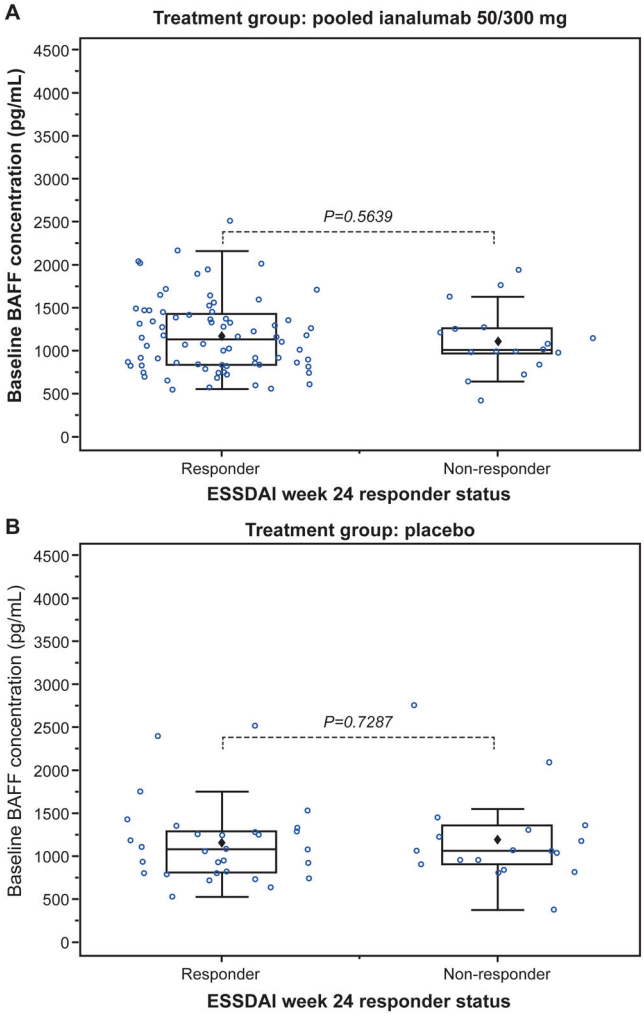

Figure 4. BAFF levels at baseline and by ESSDAI response at week 24. (A) Treatment group: pooled ianalumab 50/300 mg; (B) Treatment group: placebo. The plot shows boxes (25th–75th percentiles) with median as horizontal line. The whiskers indicate the minimum and maximum values. Each dot represents an observation. BAFF, B cell activating factor; ESSDAI, EULAR Sjögren's Syndrome Disease Activity Index.

observed up to weeks 24 and 52. As expected, B cell depletion was accompanied by a rapid increase in BAFF in a dose-dependent manner. Conversely, repopulation of B cells was paralleled by a decline in BAFF.

Assessment of the time to recovery of CD19^{+} B cells in peripheral blood is generally used to monitor the B cell compartment after B cell depletion therapy (BCDT), whereby time to recovery can vary between different autoimmune diseases.[18] In our study, the median time to recovery was dose-dependent but not dependent on pretreatment B cell counts and was comparable to experience with B cell depletion in lupus or rheumatoid arthritis.[19,20] However, our phase 2b study provided limited exposure to ianalumab for 6 to 12 months (only 300 mg dose), and no enumeration of B cell subsets was performed. Therefore, it is currently unknown whether treatment with ianalumab (or extended treatment with ianalumab at a steady state of BAFF-R engagement) might change the kinetics of B cell repopulation, the composition of the repopulated B cell compartment, or ianalumab's impact on autoantibody-producing plasma cells, memory B cells, and other cell types remains of interest.

Important factors associated with incomplete or no response to BCDT include excess BAFF[21,22] and persistence of certain B cell receptor clones.[23] It was hypothesized that serum BAFF levels reduce the threshold for autoreactive B cell survival and expansion, thereby stimulating autoantibody production. The activated BAFF/BAFF-R axis may therefore play a role in the recurrence of symptoms in rituximab-treated patients.[22,24] In a post hoc analysis, we found that high pretreatment BAFF levels were not associated with lesser responses (ESSDAI response) to ianalumab. In contrast, increased resistance to BCDT was reported in high-BAFF patients with lupus who were treated with rituximab.[19,24] These findings are consistent with the first results of trials combining treatment with BCDT and soluble BAFF blockade in SLE[25] and SjD.[15]

Significant stimulated whole salivary flow improvements were seen at week 24 for the ianalumab 300 mg group and were maintained up to the last measurement at week 48; further improvement was observed in the ianalumab 300 mg group that continued treatment versus placebo. The observed effect size of 0.45 mL/min flow improvement is clinically important as it might translate into fewer complications related to dryness, including problems in speaking and eating, emergence of oral candidiasis, and accelerated tooth decay.[26]

Exploratory efficacy results suggest that continuous dosing of 300 mg SC every 4 weeks is required for continued clinical benefit, supported by further numerical improvements in outcome measures, compared to patients who discontinued after six months of dosing and experienced worsening. Clinical improvements achieved with ianalumab 300 mg at week 24 were maintained across several end points. Due to study design limitations, no efficacy comparisons between ianalumab- and placebo-treated patients were possible after week 24. However, patients who switched from placebo to ianalumab 150 mg experienced improvements between weeks 24 and 52. Although certain PRO instruments, including PaGA, FACIT-F, and SF-36 mental component, showed consistent signals of improvement after week 24, changes in ESSPRI were small. A small further improvement (–0.95 ESSPRI points from weeks 24 to 52) was observed in patients switching from placebo to ianalumab 150 mg. A reduction of 3 points in ESSDAI and a reduction of 1 point or 15% in ESSPRI indicate clinically relevant improvement, reflecting meaningful reduction in global disease activity and symptom burden, respectively.[27] As elaborated previously,[16] the design features of this phase 2b trial may have caused a larger than expected ESSPRI placebo response, impeding the ability to

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Table 2. Summary of adverse events*

Treatment group, treatment period (duration, wk)	Ianalumab 5 mg, TP1 (24 wk), n = 47	Ianalumab 50 mg, TP1 (24 wk), n = 47	Placebo to ianalumab 150 mg,[a] TP2 (28 wk), n = 47	Ianalumab 300 mg,[b] TP1 (24 wk), n = 26,	Ianalumab 300 mg to ianalumab 300 mg, TP1, TP2 (52 wk), n = 21
Median duration of follow-up, mo	3.8	4.8	6.8	8.4	6.5
Any TEAE	43 (91.5)	43 (91.5)	45 (95.7)	25 (96.2)	20 (95.2)
Serious TEAEs	3 (6.4)	7 (14.9)	9 (19.1)	5 (19.2)	5 (23.8)
TEAEs leading to study treatment discontinuation	2 (4.3)	2 (4.3)	5 (10.6)	0 (0.0)	2 (9.5)
Deaths	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Safety topics of interest					
Local injection reaction	4 (8.5)	9 (19.1)	17 (36.2)	13 (50.0)	14 (66.7)
Systemic injection reaction	6 (12.8)	5 (10.6)	8 (17.0)	2 (7.7)	2 (9.5)
Any infections	33 (70.2)	30 (63.8)	38 (80.9)	21 (80.8)	18 (85.7)
Nasopharyngitis	7 (14.9)	4 (8.5)	11 (23.4)	5 (19.2)	4 (19.0)
Upper respiratory tract infection	7 (14.9)	9 (19.1)	7 (14.9)	8 (30.8)	2 (9.5)
Acute sinusitis	3 (6.4)	0 (0.0)	2 (4.3)	0 (0.0)	1 (4.8)
Urinary tract infection	9 (19.1)	5 (10.6)	7 (14.9)	2 (7.7)	3 (14.3)
Conjunctivitis	3 (6.4)	5 (10.6)	2 (4.3)	2 (7.7)	0 (0.0)
Gastroenteritis	1 (2.1)	2 (4.3)	1 (2.1)	2 (7.7)	1 (4.8)
Cystitis	0 (0.0)	1 (2.1)	2 (4.3)	2 (7.7)	1 (4.8)
Parotitis	2 (4.3)	0 (0.0)	3 (6.4)	1 (3.8)	0 (0.0)
Influenza	0 (0.0)	1 (2.1)	0 (0.0)	1 (3.8)	3 (14.3)
Rhinitis	1 (2.1)	0 (0.0)	1 (2.1)	1 (3.8)	1 (4.8)
Pneumonia	1 (2.1)	0 (0.0)	1 (2.1)	1 (3.8)	1 (4.8)
Tracheobronchitis	1 (2.1)	0 (0.0)	0 (0.0)	1 (3.8)	2 (9.5)
Viral upper respiratory tract infection	1 (2.1)	1 (2.1)	0 (0.0)	2 (7.7)	0 (0.0)
Blood and lymphatic disorders	8 (17.0)	10 (21.3)	6 (12.8)	4 (15.4)	5 (23.8)
Leukopenia	3 (6.4)	3 (6.4)	4 (8.5)	2 (7.7)	4 (19.0)
Lymphopenia	4 (8.5)	4 (8.5)	3 (6.4)	1 (3.8)	1 (4.8)
Neutropenia[c], n/m (%)					
<LLN–1.5 × 10^9/L (grade 1)	12/38 (31.6)	4/40 (10.0)	7/43 (16.3)	2/24 (8.3)	3/19 (15.8)
<1.5–1.0 × 10^9/L (grade 2)	9/45 (20.0)	5/45 (11.1)	6/45 (13.3)	5/25 (20.0)	2/21 (9.5)
<1.0–0.5 × 10^9/L (grade 3)	2/47 (4.3)	1/45 (2.2)	0/46 (0)	0/26 (0)	2/21 (9.5)
<0.5 × 10^9/L (grade 4)	0/47 (0)	0/45 (0)	0/46 (0)	0/26 (0)	0/21 (0)
Malignancies	1 (2.1)	0	1 (2.1)	0 (0.0)	0 (0.0)
Other common TEAEs (≥5% in any treatment group) by preferred term					
Abdominal pain	0 (0.0)	3 (6.4)	1 (2.1)	0 (0.0)	0 (0.0)
Arthralgia	3 (6.4)	1 (2.1)	7 (14.9)	2 (7.7)	3 (14.3)
Back pain	3 (6.4)	4 (8.5)	6 (12.8)	4 (15.4)	2 (9.5)
Blood creatinine	3 (6.4)	1 (2.1)	1 (2.1)	1 (3.8)	0 (0.0)
Increased					
Bronchitis	3 (6.4)	3 (6.4)	5 (10.6)	2 (7.7)	2 (9.5)
Cough	3 (6.4)	0 (0.0)	2 (4.3)	2 (7.7)	0 (0.0)
Diarrhea	3 (6.4)	3 (6.4)	2 (4.3)	3 (11.5)	4 (19.0)
Dizziness	2 (4.3)	1 (2.1)	0 (0.0)	2 (7.7)	2 (9.5)
Fall	0 (0.0)	0 (0.0)	2 (4.3)	1 (3.8)	2 (9.5)
Gastro-esophageal reflux disease	1 (2.1)	4 (8.5)	0 (0.0)	0 (0.0)	1 (4.8)
Headache	5 (10.6)	5 (10.6)	4 (8.5)	4 (15.4)	2 (9.5)
Hypertension	4 (8.5)	2 (4.3)	2 (4.3)	0 (0.0)	0 (0.0)
Influenza like illness	0 (0.0)	4 (8.5)	1 (2.1)	2 (7.7)	2 (9.5)
Injection-related reaction	6 (12.8)	5 (10.6)	8 (17.0)	2 (7.7)	2 (9.5)
Injection-site reaction	4 (8.5)	9 (19.1)	17 (36.2)	13 (50.0)	14 (66.7)
Insomnia	0 (0.0)	0 (0.0)	3 (6.4)	0 (0.0)	3 (14.3)
Iron deficiency anemia	1 (2.1)	3 (6.4)	0 (0.0)	0 (0.0)	0 (0.0)
Joint swelling	2 (4.3)	0 (0.0)	0 (0.0)	2 (7.7)	1 (4.8)
Lymphocyte count decreased	4 (8.5)	0 (0.0)	1 (2.1)	0 (0.0)	2 (9.5)
Myalgia	2 (4.3)	0 (0.0)	4 (8.5)	1 (3.8)	2 (9.5)
Neutrophil count decreased	3 (6.4)	0 (0.0)	1 (2.1)	1 (3.8)	1 (4.8)
Edema peripheral	0 (0.0)	2 (4.3)	0 (0.0)	2 (7.7)	1 (4.8)
Oral candidiasis	0 (0.0)	0 (0.0)	1 (2.1)	2 (7.7)	1 (4.8)
Oral herpes	4 (8.5)	2 (4.3)	2 (4.3)	1 (3.8)	3 (14.3)
Pain in extremity	3 (6.4)	4 (8.5)	2 (4.3)	2 (7.7)	0 (0.0)

(Continued)

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Table 2. (*Cont'd*)

Treatment group, treatment period (duration, wk)	Ianalumab 5 mg, TP1 (24 wk), n = 47	Ianalumab 50 mg, TP1 (24 wk), n = 47	Placebo to ianalumab 150 mg,[a] TP2 (28 wk), n = 47	Ianalumab 300 mg,[b] TP1 (24 wk), n = 26,	Ianalumab 300 mg to ianalumab 300 mg, TP1, TP2 (52 wk), n = 21
Pruritus	2 (4.3)	1 (2.1)	2 (4.3)	3 (11.5)	1 (4.8)
Pyrexia	0 (0.0)	1 (2.1)	3 (6.4)	3 (11.5)	0 (0.0)
Rash	6 (12.8)	3 (6.4)	3 (6.4)	2 (7.7)	2 (9.5)
Sinusitis	5 (10.6)	3 (6.4)	7 (14.9)	2 (7.7)	1 (4.8)
Sjögren's syndrome	0 (0.0)	4 (8.5)	1 (2.1)	0 (0.0)	2 (9.5)
White blood cell count decreased	3 (6.4)	1 (2.1)	0 (0.0)	1 (3.8)	1 (4.8)

* Includes all safety data from TP1, TP2, and posttreatment follow-up. Data are n (%) unless otherwise specified. A patient with multiple occurrences of an AE under one treatment is counted only once in the AE category for that treatment at the maximum severity grade; preferred terms are sorted by alphabetical order. AE, adverse event; CTCAE, Common Terminology Criteria for Adverse Events; LLN, lower limit of normal; m, number of patients with evaluable criterion who were better than the criterion at baseline; SAE, serious adverse event; TEAE, treatment-emergent adverse event; TP, treatment period.
[a] 150 mg also includes placebo-treated patients who were switched to ianalumab 150 mg at week 24.
[b] 300 mg 24 weeks group includes patients treated with 300 mg up to week 24 and switched to placebo every 4 weeks up to week 52 in TP2.
[c] Neutrophil count based on CTCAE grades.

detect patient benefit with ianalumab compared to placebo. Early and large reductions in ESSPRI scores across all arms, specifically 300 mg, may explain why only marginal further ESSPRI improvements were seen between weeks 24 and 52.

The safety events of interest include neutropenia, infections, malignancies, and injection-related systemic reactions. Hypogammaglobulinemia is known to occur because of long-term B cell depletion and may be associated with infections; hence, IgG monitoring is recommended.[28] In the current phase 2b study, safety follow-up was up to two years, and only two patients showed reductions below LLN in IgG without reports of contemporaneous infection. Likewise, a very low rate of decreased neutrophil counts was reported, and none were associated with infections except for a single incidental finding of an asymptomatic viral infection (CMV, IgM-positive). As BAFF may play a role in the development of late-onset neutropenia in patients with SLE[29] and BAFF downstream cellular effects are inhibited by the BAFF-R blocker ianalumab, it is reasonable to speculate that the combination of BCDT with BAFF-R blockade may be advantageous compared to BCDT alone. Ongoing larger trials will shed further light on these relationships.

Infections were seen more frequently with ianalumab compared to placebo, and events included bacterial and viral infections. No severe or systemic opportunistic infections were observed in this phase 2 trial. Patients with SjD generally have fewer background immunosuppressive treatments compared to patients with SLE. Ongoing phase 3 trials will provide additional insights into potential events that occur rarely under BCDT.

The limitations of this study include the small sample size after week 24 for the 300 mg subgroup comparisons, which were descriptive only and not powered for statistical significance, and no collection of efficacy data for the lower doses of 5 and 50 mg after the primary endpoint at week 24. The placebo group in TP2 was not designed to control for efficacy; therefore formal statistical comparisons between treatment groups during TP2 was not feasible.

In conclusion, week 52 results of ianalumab in SjD, including up to two years of safety follow-up, yielded a positive risk-benefit profile and provided additional insights into the kinetics of biomarkers during treatment and after treatment discontinuation. The findings suggest that ianalumab treatment should be administered for longer than six months because continued treatment resulted in additional benefit and no new safety concerns, whereas discontinuation after six months (ie, shortly after steady state is achieved) resulted in rebound of disease activity in some patients.

AUTHOR CONTRIBUTIONS

All authors contributed to at least one of the following manuscript preparation roles: conceptualization AND/OR methodology, software, investigation, formal analysis, data curation, visualization, and validation AND drafting or reviewing/editing the final draft. As corresponding author, Dr Dörner confirms that all authors have provided the final approval of the version to be published and takes responsibility for the affirmations regarding article submission (eg, not under consideration by another journal), the integrity of the data presented, and the statements regarding compliance with institutional review board/Declaration of Helsinki requirements.

ROLE OF THE STUDY SPONSOR

The study was designed by the authors including Novartis personnel. All authors had access to the data, contributed to the interpretation, and collaborated in the development of the manuscript. Statistical analyses were performed by statisticians employed by the study sponsor (Novartis Pharma AG, Basel, Switzerland). Publication of this article was not contingent upon approval by Novartis. Medical writing and editorial assistance were provided by Divya Chandrasekhar, PhD (Novartis Healthcare Pvt. Ltd., India), and Ramji Narayanan, M Pharm (Novartis Ireland Limited), which was funded by Novartis Pharma AG.

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DATA AVAILABILITY STATEMENT

All data relevant to the study are included in the article or uploaded as supplementary information. Novartis is committed to sharing, with qualified external researchers, access to patient-level data and supporting clinical documents from eligible trials. These requests are reviewed and approved by an independent review panel on the basis of scientific merit. All data provided are anonymised to respect the privacy of patients who have participated in the trials in line with applicable laws and regulations.

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